

The Future of Interaction with Mobile Game Characters

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Figure 1: Verbal interactions with the NPC. Subfigure (a) show it holding a consistent conversation, (b) using a Multilayer Perceptron (MLP) with Unity ML-Agents to throw a basketball, (c) using MLP with Unity ML-Agents to push the chair.

ABSTRACT

This poster showcases the future of player-NPC interaction in gaming: more natural, based on speech. The poster explains the implementation of verbal interaction with an NPC in a mobile game. Large Language Models (LLMs) open new ways of interacting in gaming, but the large size and memory footprint make deploying them on mobile a challenge. We have identified a 33M-parameter small Language Model (LM), TinyStories [Eldan and Li 2023], repurposed it to be conversational, and combined it with the standard Unity Multi-Layer Perceptron (MLP) ML-Agents [Juliani et al. 2020] model. The LM drives the interaction with the user, while the MLP performs the actions. A Sentence Similarity model [Reimers and Gurevych 2019] is then responsible for filtering user input to decide whether the NPC needs to communicate with the player or to perform an action. Running all these models locally on mobile inside a game engine like Unity can be very challenging in terms of optimization, but this poster demonstrates how it is possible.

CCS CONCEPTS

• **Computing methodologies** → **Speech recognition**; • **Human-centered computing** → **Smartphones**.

KEYWORDS

user interaction in games, speech-based user interaction, speech recognition, language model, unity ml-agents, mobile games

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1 INTRODUCTION

After a few years working with Unity ML-Agents [Mitsunami 2023] we realized the limitations of the simple MLP brain the agents use. It is great in terms of using Reinforcement Learning (RL) to perform actions, nevertheless it only provides support for indirect modes of player-NPC interaction. At some point, we understood that the MLP model needed to be replaced or complemented with a model suitable for more direct interaction modes. The most obvious candidate was a language model (LM), but the performance can be an issue because of the need to run the model inside a game engine on a smartphone at a latency suitable for real-time gaming. Low memory footprint is also a strong requirement, considering that an average smartphone in 2024 comes with 8 GB of RAM or less.

Unfortunately, currently available LMs are far from mobile-friendly, with parameters that are upwards of billions in size. However, we were fortunate that the TinyStories research paper helped relieve this issue. It provided us with access to a 33M-parameter LM with only four transformer blocks, making inference in real-time while maintaining a low memory footprint a possibility. We decided to keep the ML-Agents MLP, meanwhile adding an LM to take care of the interaction with the layer and communicate with the MLP model. In this way, we keep the agent's ability to perform action, and extends it with verbal interactivity. This approach introduces our view of the future of player interaction in games based on natural speech.

2 IMPLEMENTATION

2.1 The Language Model

A key component in the implementation of the idea is the TinyStories LM. This model came about as a result of an investigation to determine the smallest language model that still possesses coherent conversational and reasoning capabilities. The resulting 33M-parameter model is equivalent to the brain of a 3-4 year-old child. As the model was trained on a dataset consisting of mainly short stories, it needed to be repurposed to enable conversational interactions, which involved various prompting techniques.

In addition, we needed a speech-to-text (STT) model to convert the user speech to text, to feed the LM. A second text-to-speech (TTS) model was also needed to convert the output of the LM to speech. It seemed like an easy task to find such models, but models that provide good quality and performance on mobile proved difficult to find. STT and TTS models require heavy pre- and post-processing, and, in most cases, they are not integrated with the main model. After careful consideration, we opted for the native Android built-in STT and TTS APIs, which proved to be more compact and optimized, making it more suitable for our demonstration.

Finally, we make use of a Sentence Similarity (SS) model to identify if in the context of the user-NPC interaction, there is reference to any of the actions the ML-Agent can perform.

2.2 The Execution Pipeline

The LM models used in the user-NPC interaction and the MLP model of the ML-Agent are running on the Unity inference engine known as Sentic, except for the STT and TTS models. These models are managed by the Android system.

All the models in this work are accelerated on the CPU. The NPC's MLP model has been trained using RL to perform several complex physics-based actions: pick a basketball and throw it into the hoop, move a chair to reach a car etc. The RL training has been performed in the Unity ML-Agents environment.

The application runs on a Google Pixel™ 8 smartphone (1x3.0 GHz Arm® Cortex®-X3 & 4x2.45 GHz Cortex®-A715 & 4x2.15 GHz Cortex®-A510 CPU cores). See the screen captures from the app running on the mobile device in Figure 1. The sequence of model execution is represented in Figure 2.

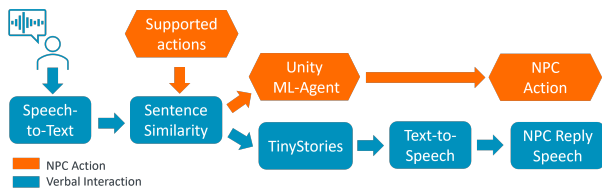


Figure 2: System diagram showing the full pipeline.

The table below shows the execution time for each part of the pipeline, measured in developer mode. This mode has been implemented to profile the execution of different parts of the app pipeline.

Table 1: Average Initial Reaction Time

Stage	Action(ms)	Dialogue (ms)
Sentence Similarity	30.0	25.2
TinyStories 33M	-	59.7
Misc.	2.5	35
Overall Reaction	32.5	119.9

The performance evaluation reveals that the TinyStories LM and the Sentence Similarity together are still the bottleneck of the system in terms of execution speeds. However, the overall latency (reaction time) is still more than satisfactory for real-time gaming.

3 CONCLUSIONS

LLMs are being increasingly adopted, and using speech as input results in a more natural way of interacting, which helps democratize the use of LLMs like ChatGPT or Copilot. This poster emphasizes the growing significance of language models in gaming and technology. It explores new ways of utilizing Unity ML-Agent Toolkit to create complex NPCs capable of performing actions and interacting verbally with the player. It shows what is coming next in gaming: natural speech interaction between users and even smarter NPCs powered by LLMs, which makes the game experience more enjoyable, immersive, and richer.

We consider verbal interaction in games as a first step toward wider use of LLMs in games. LLMs will open the possibility to implement an LLM-based omnipresent AI companion capable of helping and learn our behavior during the game play. While in this work we make use of a small LM, there are ongoing efforts to optimize LLMs inside game engines using low-level specialized libraries like KleidAI [Iodice 2024] for Unity Sentic on Arm® architectures. The poster demonstrates that the current technology is almost there to achieve this on mobile.

Verbal interaction humanizes and enlivens NPCs, enriching the interactive gaming experience and making it accessible to wider audiences. We hope this poster inspires the development of more compact, mobile friendly LMs.

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